



Wyoming operating geography and engineering derivation

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Record origin: User-authorized fictional case implementation, derived from controlled handoff and explicitly labeled engineering assumptions. Official sources establish external context only; they do not verify fictional assets, transactions, operating events or professional approvals. References observed September 6, 2026 UTC / September 5 PDT unless the pinned source manifest records an earlier acquisition.

Taylor is a fictional industrial town at **42.120000° N, 108.100000° W**. The Wamsutter attachment point is **41.6744965784° N, 107.9656271521° W**, selected against official FRA rail reference feature REF-WAMSUTTER-FRA-RAIL-14, whose source owner field is UP. The attachment and operating agreement are fictional case facts; the reference geometry is not permission to modify an actual railroad. Red Wash is the separately owned mine at **42.220000° N, 108.180000° W**, in Sweetwater County.

The prior geospatial branch supplied three screened Taylor candidates. Candidate A's mainline measures 33.3484768 geodesic miles, leaving 6.6515232 miles for the independent customer branches within the 40-mile system. Candidates B and C would consume about 37.0615 and 38.8133 miles before branches. A was therefore selected because it leaves room for the inherited logistics estate and does not turn the system into a near-direct mine branch. No mine rail-connector candidate was adopted.

The East Materials Branch joins the main at MP 15 and extends four miles northeast. The Mineral Transfer Branch joins at MP 26 and extends 2.6515232 miles northeast. These are derived historical external-customer lines, with opening dates of May 8, 1972 and September 19, 1986. The 1898 coal origin, 1953 parent failure and July 1, 1954 rescue remain distinct from the derived October 14, 1968 completion of the longer Taylor mainline. Parallel classification, shop, passing and terminal tracks do not add route-miles.

The truck-only mine access road measures nine geodesic miles. Its existing industrial-road status is a synthetic case assumption; it is not a claimed identified public road or land right. The \$450,000 mine road allocation is targeted rehabilitation of that assumed existing road, not the cost of building a new nine-mile road. Seasonal drainage and maintenance remain operating obligations. The road model does not reuse the old map's WYO130 access statement.

The source GeoJSON uses EPSG:4326 longitude/latitude. The new mine anchor transforms to approximately **732,748.782 m easting / 4,678,053.611 m northing in NAD83 / UTM 12N**. The old Carbon County point lay in standard UTM zone 13. Regional output maps use a clearly declared local equirectangular metric frame for screening; neither that declaration nor the local mine schematic relabels an old UTM coordinate. The retained DEM remains in its original EPSG:26913 projection, with proper transformations used to sample it.

Pinned USGS 3DEP data gives a bilinear anchor screening elevation of **2,098.424 m, approximately 6,885 feet**. This replaces the unsupported old 6,420-foot annotation. Resolution is about 70.211 metres; these elevations do not substitute for a survey.

The inherited mainline profile permits a maximum 1.8% conceptual formation grade, with about 6.87 m maximum cut and 5.89 m fill. Its sampled plan curvature minimum is about 701 m. New branch screening gives East Materials about 4.92 m maximum cut and 4.55 m fill at the 1.8% formation constraint. Mineral Transfer's sampled ground grade is about 1.36%, needing no grade correction in this screen. The nine-mile road's maximum sampled ground grade is about 3.80%, below the declared 6% road envelope. These are grade-only or preliminary formation screens; they do

not establish vertical curves, cross-sections, earthwork quantities, soil stability, bridge hydraulics or construction cost. Independent spatial checking rejected an initial Mineral branch that crossed roughly 206 m of mapped waterbody and an East alternative that crossed roughly 45 m. The final mainline, both branches and road have **zero overlap with the pinned waterbody polygons**. That is narrower than a wetlands or stream-clearance finding. Ephemeral drainage structures remain in the synthetic inventory. The two actual I-80 reference intersections occur at projected mainline mileposts about 0.4043 and 0.4338; synthetic highway overpasses are placed there. The initially assumed MP 0.72 crossing was discarded. No undisclosed new major railroad bridge is used to make the line work.

source/geography/reference_manifest.json preserves the URLs, retrieval dates, public-domain attributions and source hashes of the reused Census/FRA/USGS layers. dem_manifest.json records the USGS request and exact raster hash. candidate_comparison.json, derived_ground_profiles.json and spatial_screen.json retain the numerical evidence and limitations. Only selected geospatial source/reference files were reused; no release archive, execution journal or duplicate repository snapshot was merged.

Current official context, retrieved September 6, 2026 UTC / September 5 PDT: [USGS UTM explanation](#), [WSGS regional uranium deposits](#), [USGS 3DEP service](#). Regional uranium occurrence does not certify the fictional underground resource or mining method.